

Minor Project Report

Smart Dustbin – Physical Built

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1. Introduction

The objective of this project is to develop an automated waste management system that combines hands-free lid opening via proximity sensing with real-time container capacity tracking. Additionally, the system incorporates firmware-level monitoring to calculate and log estimated energy consumption across operational states.

2. Hardware Architecture and Pin Configuration

The smart dustbin is implemented using an ESP32 DevKit V1 as the main controller. Two HC-SR04 ultrasonic sensors are used for fill-level measurement and hand/proximity detection. An SG90 servo controls the lid, while an active buzzer provides an alert when the bin reaches the defined capacity threshold. A 16×2 I2C LCD provides real-time status information.

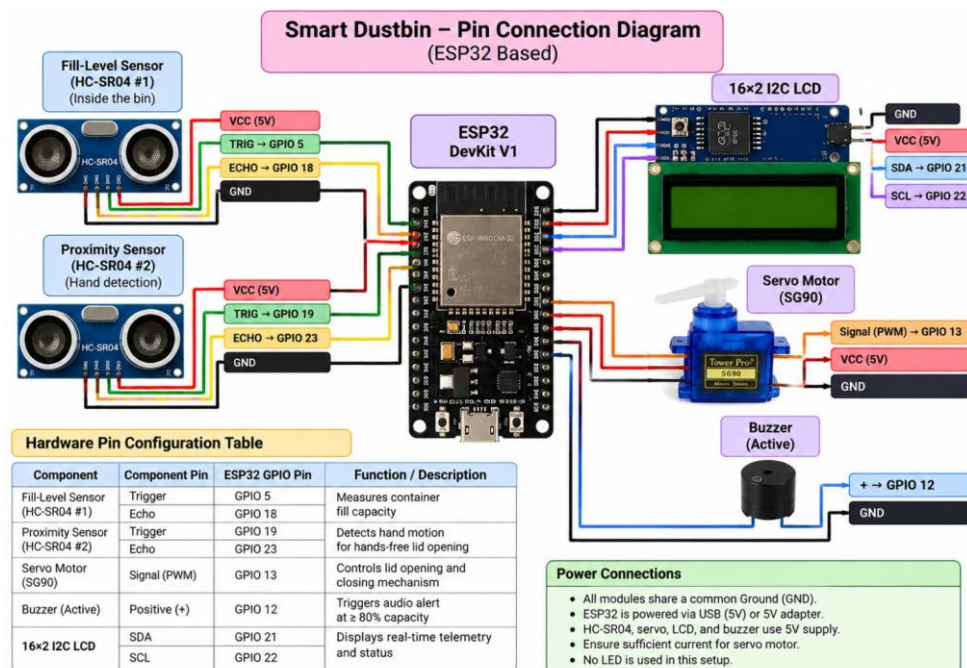


Figure 1: Complete hardware pin connection diagram and configuration table for the ESP32-based smart dustbin.

3. Core Implementation and Software Architecture

The firmware is built using modular state-machine logic to ensure efficient, non-blocking execution.

- Key Concepts Applied: Pulse-Width Modulation (PWM) for precise servo motor control, I2C communication protocol for the LCD, non-blocking timing using `millis()`, and conditional state logic.

Program Logic Flow:

- Initialize all GPIO pins, serial communication, and the I2C LCD.
- Continuously read the fill-level sensor to compute the bin's fill percentage against a calibrated 12 cm empty baseline.
- Evaluate the fill level against the safety threshold ($\geq 80\%$), triggering the warning buzzer if reached (ALERT STATE).
- If capacity is normal, monitor the proximity sensor; open the lid via servo motor when a hand is detected within range (SENSING STATE).
- Track operational durations in milliseconds for IDLE, SENSING, and ALERT states to calculate cumulative energy drain.
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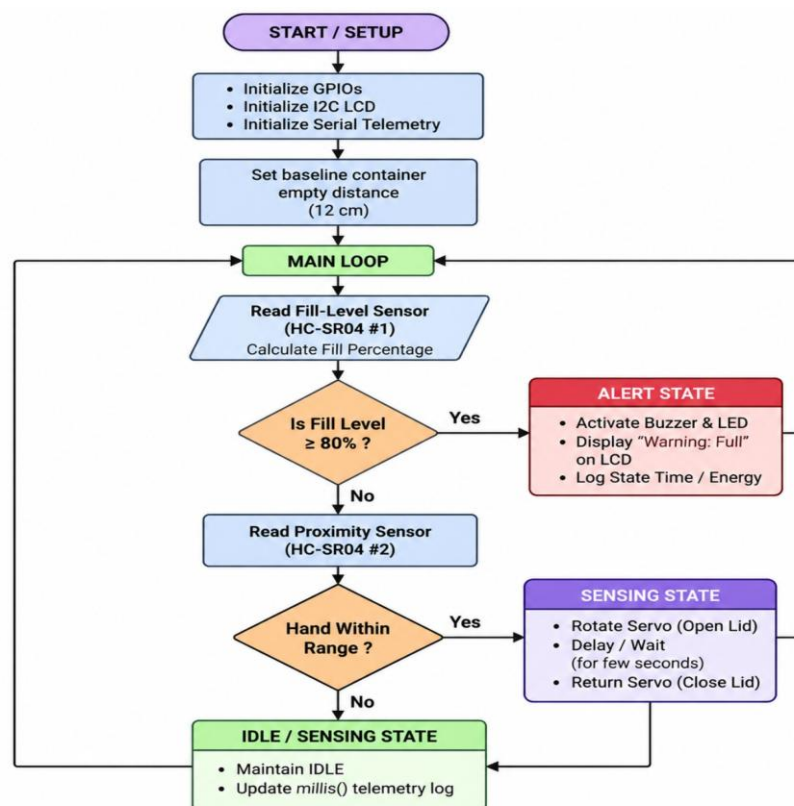


Figure 2: Software flowchart illustrating system initialization, main loop monitoring, and conditional state logic (IDLE, SENSING, ALERT).

4. Results, Telemetry, and Energy Analysis

System performance was verified through physical interaction trials, real-time LCD status checks, and live serial monitor telemetry logging.

Power and Energy Drain Breakdown

- IDLE State: 278 s duration | 40.00 mA current draw | 0.200 W power dissipation
- SENSING State: 6 s duration | 150.00 mA current draw | 0.750 W power dissipation
- ALERT State: 13 s duration | 80.00 mA current draw | 0.400 W power dissipation
- Total Estimated Energy Drain: 3.6651 mAh

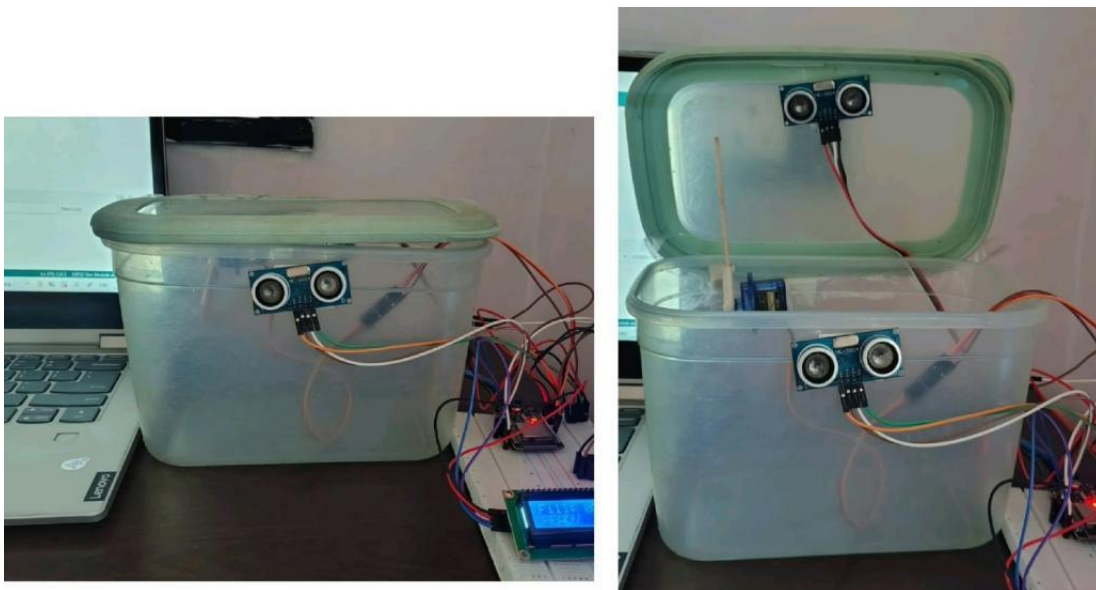


Figure 3: Physical prototype setup showing the container assembly, ultrasonic sensors, and I2C LCD status display.

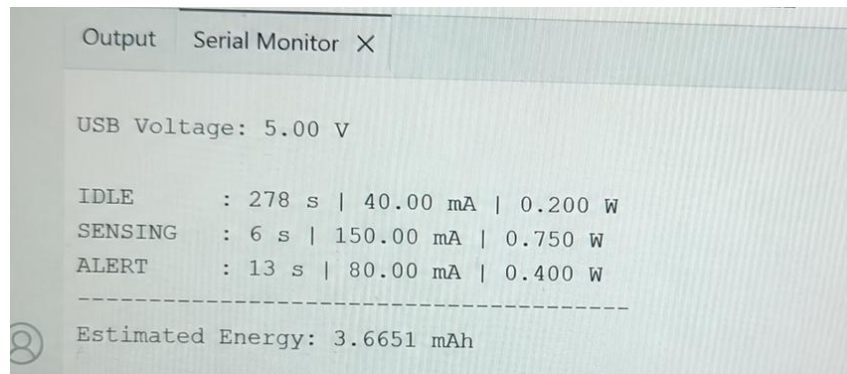


Figure 4: Live serial monitor telemetry showing state durations, current draw, power dissipation, and cumulative energy consumption.

5. Conclusion and Engineering Challenges

- **What I Learned:** This project provided practical, hands-on experience in interfacing multiple ultrasonic sensors with an ESP32 Dev Kit and translating state-machine timelines into quantifiable power metrics.
- **Technical Challenges & Solutions:** A primary challenge encountered during prototyping was false readings caused by mounting the fill-level sensor on the moving lid. This was successfully resolved by relocating the sensor to a fixed enclosure surface and updating the baseline calibration parameters.